

**Behavior anomaly detection using machine learning to ensure
continuous authentication**

Project ID: 25-26J-072

Individual Project Proposal Report

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BSc. (Hons) in Information Technology Specializing in Cyber Security

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DECLARATION OF THE CANDIDATE AND SUPERVISOR

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Abstract

In Local Area Network (LAN) environments, normal access control methods primarily target security of the authentication step during login, frequently overlooking the risks posed by post-login hijacking attacks. This mistake renders systems susceptible to vulnerabilities like physical hijacking and remote session hijacking, wherein adversaries exploit active sessions to obtain unauthorized access. To address this challenge, this research proposes the development and evaluation of a machine learning-based behavioral anomaly detection system designed to ensure continuous authentication throughout a user's session. The system seeks to detect real-time anomalies indicative of potential hijacking attempts by analyzing multi-context behavioral patterns, such as keyboard dynamics, mouse movements, and system interaction aspects. The primary objective is the construction, training, and optimization of machine learning models proficient in differentiating between genuine and anomalous user behaviors with high precision, while concurrently reducing false positives. This technique introduces a session-level, continuous authentication mechanism specifically designed to address post-login hijacking attacks, a domain often neglected in traditional LAN security strategies. This work enhances LAN file system security by offering adaptive, intelligent safeguards against insider misuse and hijacking threats, thereby maintaining trust, accountability, and resilience in secure file access contexts.

Keywords: -LAN Security, Continuous Authentication, Behavioral Anomaly Detection, Machine Learning, Post login Hijacking, Physical Hijacking, Remote Session Hijacking, User Behavior Analytics ,Adaptive Access Control, Session-level Security.

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1. Introduction

Traditional authentication mechanisms in Local Area Network (LAN) environments, such as passwords, tokens, and biometric verification, are primarily focused on granting initial access to the system. However, once a user is authenticated, these mechanisms rarely validate the identity of the active session continuously. This limitation exposes systems to post-login hijacking threats, including physical hijacking and remote session hijacking, where an attacker takes control of an authenticated session without triggering alerts. To overcome these challenges, continuous authentication methods have emerged, relying on user behavior monitoring to provide ongoing verification. Machine learning (ML)-based behavioral anomaly detection offers a promising solution by analyzing user interaction patterns and identifying deviations that may indicate malicious activity.

Background

Authentication has always been fundamental to safeguarding digital systems, especially in Local Area Network (LAN) contexts where sensitive data is often exchanged among numerous users and endpoints. Traditional authentication methods, including passwords, PINs, smart cards, and biometric verification, are effective at the time of login; but they do not ensure that the authenticated user stays the legitimate individual during the session. Once an assailant acquires access to an authenticated session—either through physical hijacking (e.g., seizing control of an unlocked workstation) or remote session hijacking (e.g., session token theft, man-in-the-middle attacks)—the system typically lacks ways to re-validate user's identity.

This difficulty underscores the necessity for continuous authentication—a security framework that consistently verifies the user's identity throughout the session. In contrast to conventional one-time verification, continuous authentication depends on behavioral biometrics and contextual indicators, including keyboard dynamics, mouse movement patterns, application usage behaviors, and system interaction records. These signals are often inconspicuous and can be observed in the background without interfering with user engagement.

The advancement of machine learning (ML) and anomaly detection methodologies has enhanced the viability of continuous authentication. Machine learning algorithms can assimilate the typical behavioral profile of an authentic user and identify nuanced variations that may indicate impersonation or malicious control. Supervised techniques (e.g., Random Forest, Support Vector Machines, Neural Networks) and unsupervised techniques (e.g., Autoencoders, One-Class SVM, Isolation Forest) have demonstrated significant efficacy in modelling user behavior. Furthermore, hybrid models that incorporate multi-modal data (keystroke, mouse, and context-aware features) offer enhanced authentication by offsetting discrepancies in individual behavioral characteristics.

In LAN file systems, where insider attacks and session hijacking can lead to significant breaches of confidentiality, integrity, and availability, machine learning-based behavioral anomaly detection provides a scalable, adaptable, and proactive defiance strategy. It facilitates the detection of unauthorized access post-login and enables real-time mitigation measures—such as automatic session termination, re-authentication prompts, or access restrictions—upon the identification of anomalies. [1]

Literature Survey

A number of studies in recent years have contributed to the field of behavioral anomaly detection and continuous authentication:

1. **Mondal & Bours (2017)** conducted a comprehensive survey of continuous authentication systems based on behavioral biometrics. Their work highlighted keystroke and mouse dynamics as effective modalities, achieving detection accuracy above 90% in controlled experiments.
2. **Ehatisham-ul-Haq et al. (2018)** proposed a continuous authentication model using smartphone sensor data (motion, location, and app usage). Their results showed that multi-context behavioral features improved anomaly detection compared to unimodal approaches, demonstrating the value of feature fusion.
3. **Alshehri & Hussain (2020)** explored ML-based anomaly detection in enterprise networks, emphasizing session-level monitoring. Their framework adapted to user activity patterns in real time, achieving a significant reduction in false positives compared to rule-based methods.
4. **Zheng et al. (2019)** introduced a session anomaly detection mechanism for desktop environments, combining keystroke and mouse dynamics. Their system effectively detected hijacking attempts with minimal user disruption, which is closely aligned with the objectives of this research.
5. **Patel et al. (2021)** developed a continuous authentication system using deep learning to analyze fine-grained keystroke dynamics. They demonstrated that Recurrent Neural Networks (RNNs) and Convolutional Neural Networks (CNNs) outperform traditional ML classifiers in modeling temporal behavioral data.
6. **Alzubaidi & Kalita (2016)** emphasized that behavioral biometrics are resilient against theft or replication compared to static authentication methods like passwords, further strengthening the argument for continuous verification.
7. **Garg & Batra (2022)** proposed a hybrid continuous authentication model that integrated behavioral features with contextual metadata (e.g., login time, network attributes). This

hybrid approach reduced the chances of false positives caused by behavioral variability under stress or fatigue.

8. **Recent advancements (2022–2024)** also indicate a growing shift towards **adaptive and explainable AI (XAI) techniques** in anomaly detection for authentication. For example, SHAP-based interpretations allow administrators to understand why a model flagged a session as anomalous, increasing trust in ML-driven security systems.

Despite these advancements, most existing studies focus on mobile devices or general computing environments. **Very few address LAN-specific contexts**, where **file system access and session hijacking threats** are prevalent. Moreover, the majority of continuous authentication systems are evaluated on controlled datasets, lacking real-world deployment in enterprise or organizational LANs. This research aims to fill that gap by implementing a **multi-context, ML-driven continuous authentication system for LAN environments**, specifically targeting **physical and remote session hijacking attacks**. [2]

Research Gap

Research	Focus on LAN environments	Continuous authentication	Post-login hijacking detection	Multi-context behavioral features	Machine Learning-based anomaly detection
Research A [1]	X	✓	X	X	✓
Research B [2]	X	✓	X	✓	X
Research C [3]	X	X	✓	X	✓
Research D [4]	X	✓	X	✓	✓
Proposed Work (SafeLAN – Behavioral Anomaly Detection)	✓	✓	✓	✓	✓

Research Problem

Local Area Networks (LANs) are the backbone of modern digital systems. They make it easy for users to share files and communicate with each other securely. The use of static, one-time security methods like passwords, smart cards, or biometrics, on the other hand, makes systems much more vulnerable. Once users are given access, they are trusted for the whole session and don't have to prove who they are again. Because of this hole, systems can be attacked after the user logs in. These attacks include physical hijacking (where an attacker takes over an unattended, logged-in workstation) and remote session hijacking (where attackers capture or use authenticated sessions). Attacks like these let bad guys get around normal access controls and work inside a trusted session without being seen. This can lead to data theft, insider abuse, and breaches of LAN file systems' confidentiality, integrity, and availability.

This problem encapsulates several technical challenges:

- Identifying suitable **behavioral and contextual features** (e.g., keystroke patterns, mouse dynamics, system usage logs) that are discriminative yet non-intrusive.
- Developing and evaluating **machine learning models** capable of accurately modeling user behavior while adapting to natural variations over time.
- Ensuring **real-time anomaly detection** that can trigger timely mitigation strategies (session lockout, re-authentication prompts) without degrading user experience.
- Addressing the lack of **LAN-specific deployment frameworks**, which are essential for securing enterprise file systems against insider threats and hijacking attempts. [3] [4]

2. OBJECTIVE

Main Objective

The main objective of this research is to design, execute, and assess a machine learning-based behavioral anomaly detection system that facilitates continuous authentication in LAN environments, specifically targeting the identification and mitigation of post-login hijacking threats, including physical and remote session hijacking. This purpose aims to rectify the deficiencies of conventional one-time authentication methods by implementing a multi-context, session-level security architecture that continuously verifies the authenticity of an ongoing session. Utilizing behavioral biometrics (including keystroke dynamics, mouse movement patterns, and system interaction features) alongside advanced algorithms for machine learning, the system will continually monitor user behavior to establish a baseline of typical activity and identify anomalies that may suggest an unauthorized intrusion. The primary objective is to establish a non-intrusive, adaptive, and real-time security solution that enhances LAN file system protection against insider threats and hijacking attempts, while reducing false positives and ensuring a seamless user experience.

Specific Objectives

1. **To identify and select appropriate behavioral and contextual features** (e.g., keystroke dynamics, mouse movement patterns, system usage behavior) that can be effectively used for modeling user activity in LAN environments.
2. **To design and develop machine learning models** capable of learning normal user behavior and detecting anomalies that may indicate physical or remote session hijacking attacks.
3. **To implement a session-level continuous authentication framework** that integrates anomaly detection into the LAN file system, ensuring seamless and non-intrusive user verification during active sessions.
4. **To evaluate and optimize the performance** of the proposed system using relevant metrics such as accuracy, precision, recall, F1-score, and false positive rate, ensuring reliability in real-world LAN environments.
5. **To propose mitigation mechanisms** (e.g., automatic session lockout, re-authentication prompts, or access restrictions) that are triggered upon detection of suspicious behavior, minimizing security risks while preserving usability. [5]

3. Methodology

The methodology of this research focuses on the design, development, and evaluation of a machine learning–based behavioral anomaly detection system to enable continuous authentication in LAN environments. The process will be structured into the following phases:

1. Requirement Analysis and Problem Identification

Identify the principal security concerns in LAN environments, including physical hijacking and remote session hijacking. Establish system requirements for continuous authentication, encompassing security, scalability, usability, and real-time detection capabilities. Define the assessment criteria and performance standards (accuracy, precision, recall, false positive rate, F1-score).

2. Feature Identification and Data Collection

Determine contextual and behavioral characteristics, such as typing speed, dwell duration, and flight time, that indicate user engagement during LAN sessions. Mouse movements are behavior, including scroll patterns, click frequency, and trajectory. Patterns of system usage: command execution, file usage frequency, and application access. Create or model a LAN-based testing environment to gather behavioral data from numerous users in real time. To get the gathered data ready for machine learning, preprocess it using normalization, feature extraction, and noise reduction.

3. Machine Learning Model Development

Examine various machine learning techniques appropriate for anomaly detection, such as: Support Vector Machines (SVM), Random Forest, and Neural Networks are examples of supervised learning. Autoencoders, One-Class SVM, and Isolation Forest are examples of unsupervised learning. For increased resilience, hybrid techniques combine supervised and unstructured models. Use data on typical user behavior to train models, then assess how well they can identify anomalies that indicate hijacking attempts. Use multi-context learning, which combines several behavioral traits to improve detection precision and lower false positives.

4. Continuous Authentication Framework Design

Create session-level monitoring framework to gather behavioral data during user sessions. Utilize real-time anomaly detection to match user behavior with the established profile. Integrate anomaly classification decision mechanisms. Continue session if normal. If an anomaly is found, implement mitigation measures such as session lockout, step-up authentication, and limited access. Ensure system usability and user acceptance by minimizing intrusions.

5. System Implementation in LAN Environment

Roll out the system prototype in a supervised local area network (LAN), such as a lab-based file sharing system. Play out potential assaults, such as:

- o A physical hijacking, in which the attacker takes control of an unprotected session.

Attacks that hijack a remote session, such as token or session replay attacks.

Keep an eye on how the system reacts and make sure it can detect anomalies in real-life LAN scenarios.

6. Performance Evaluation

Assess system performance using the following metrics:

- o Detection Accuracy (accurately detecting attempts at hijacking).
- False Positive Rate (FPR): when legitimate users are mistakenly reported.
- F1-score (efficiency of anomaly categorisation), precision, and recall.
- Response Time (the capacity to identify and mitigate in real time).

Examine outcomes in relation to baseline techniques (such as static access restriction or unimodal continuous authentication).

To evaluate the effect on the user experience, perform usability testing.

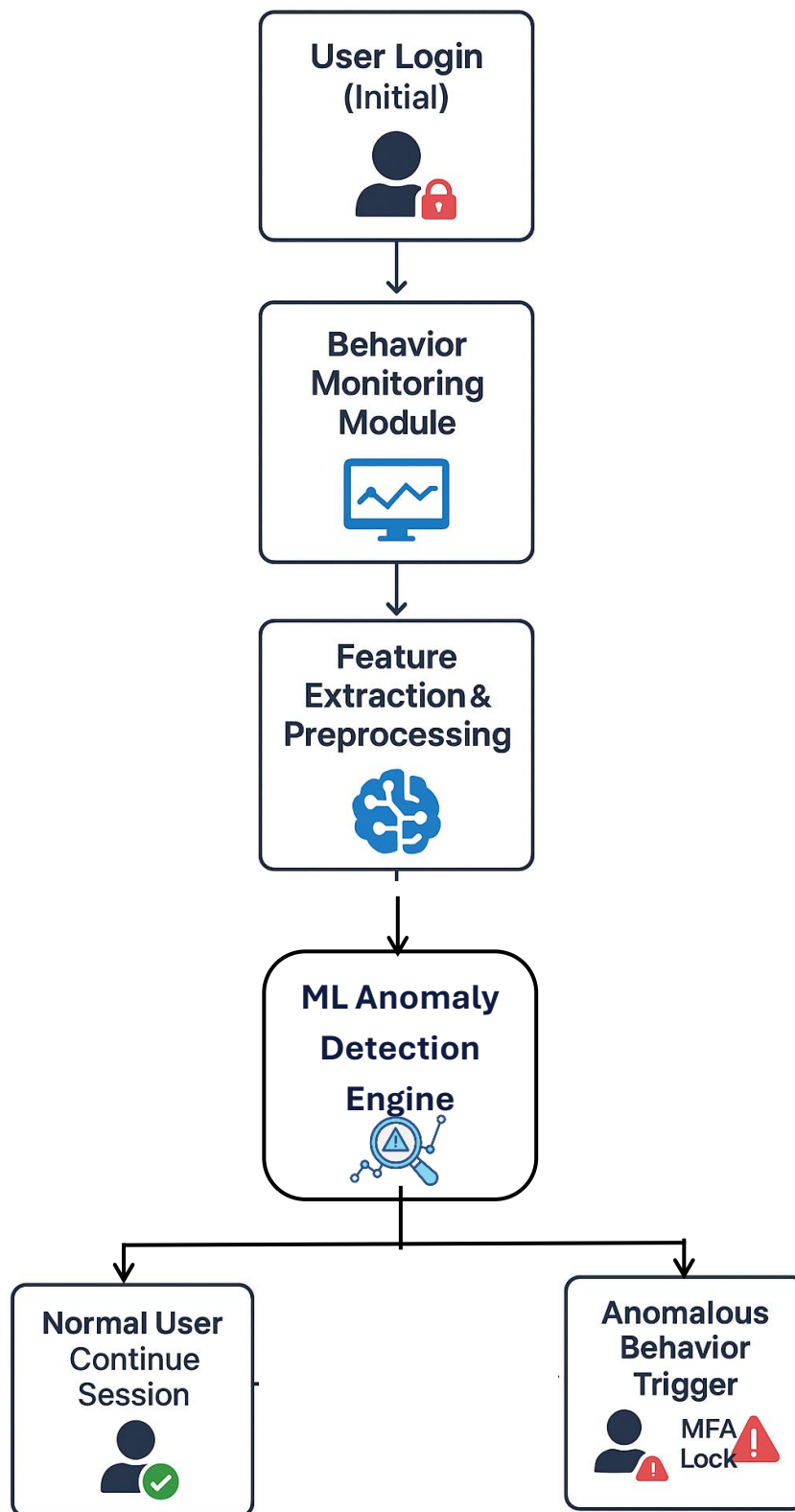
7. Optimization and Refinement

Optimize feature selection and model hyperparameters to achieve a balance between security and usability. Explore adaptive learning methodologies, such as online learning and reinforcement learning, to perpetually refine behavioral models in response to the evolution of user behavior. Propose enhancements to bolster resilience against adversarial attacks, in which attackers seek to replicate user behavior.

8. Documentation and Knowledge Contribution

Document system architecture, data processing pipeline, ML models, and evaluation results. Demonstrate how ML-driven multi-context continuous authentication addresses post-login hijacking detection in LAN file systems, answering a research gap. Offer a path for realistic enterprise LAN adoption. [6] [7] [8]

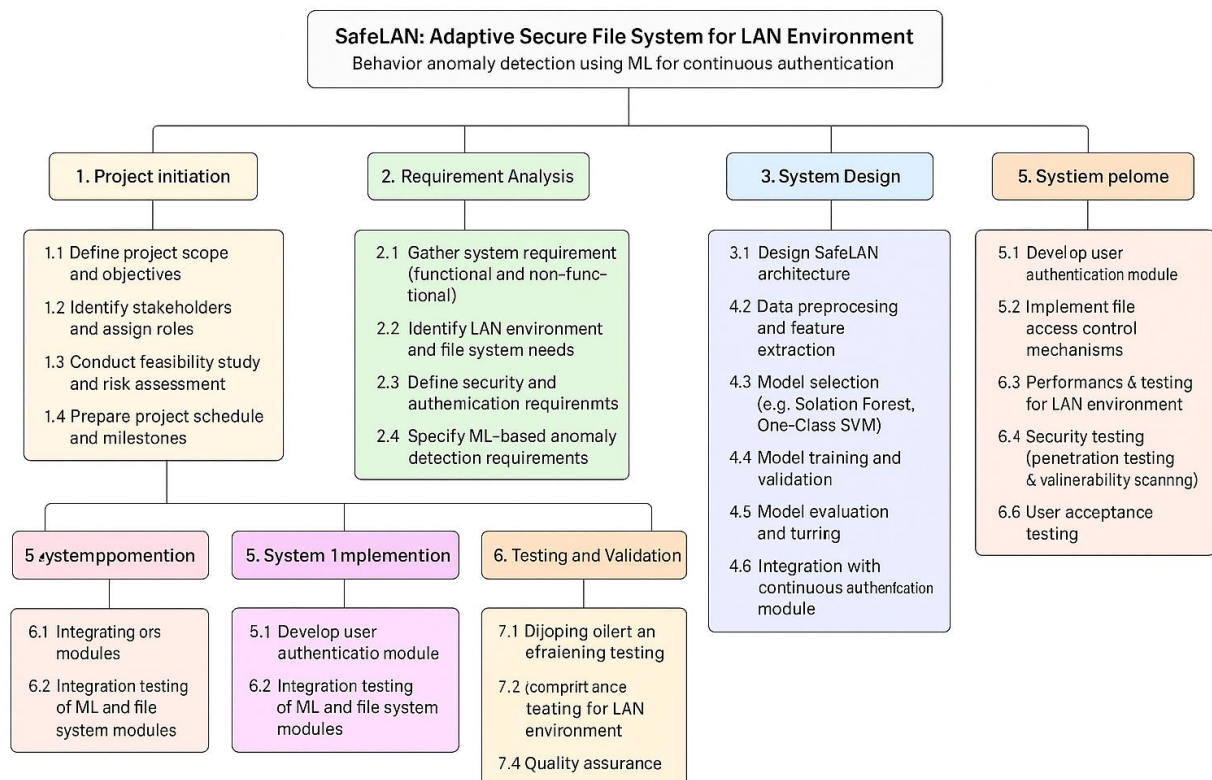
System Diagram



System Architecture for Safe LAN – Continuous Authentication

1. **User Interaction Layer:** Users may gain access to the LAN file system by using a normal login method, like a username and password or a smart card. The session action starts after the first authentication.
2. **Behavior Monitoring Module:** Collect continuous behavioral data: Keystroke dynamics, Mouse movements, Application/file access patterns, Network activity logs.
3. **Feature Extraction & Preprocessing:** converts unstructured behavioral data into organized feature sets. Implements normalization, noise reduction, and dimensionality reduction techniques.
4. **Machine Learning–based Anomaly Detection Engine:** Pre-trained ML models compare live behavior with user’s normal profile. Models One-Class SVM, Autoencoder, or Hybrid ML
Output: Normal vs. Anomalous activity.
5. **Decision & Response Manager**
 - If behavior matches profile → continue session.
 - If anomaly detected →
 - Trigger **secondary authentication** (MFA prompt).
 - Lock session / restrict access to files.
 - Alert system admin for investigation.
6. **Secure LAN File System:** Requires continuous authentication for access. Records anomalies and answers in security audit trail.

WORK BREAKDOWN CHART



4. Project Requirements

1. Functional Requirements

1. **User Authentication**=Provide initial login authentication (username/password, biometric, or smart card).Ensure seamless integration with LAN-based file access systems.
2. **Behavior Monitoring**=Continuously monitor user interactions such as keystrokes, mouse dynamics, file access patterns, and session activities. Capture data in real time with minimal performance overhead.
3. **Feature Extraction & Preprocessing**=Draw out pertinent behavioral characteristics from the information gathered. Preprocess and normalize the data before entering it into machine learning models.
4. **Machine Learning–Based Anomaly Detection**=Develop and implement anomaly detection models (e.g., One-Class SVM, Random Forest, Autoencoders). Contrast real-time user behavior with predefined behavioral profiles. Output classification: Normal session versus Anomalous session.
5. **Continuous Authentication**=Authenticate users throughout their session, not just at login. Detect and mitigate threats such as:
 - **Physical Hijacking** (e.g., someone taking over an unlocked workstation).
 - **Remote Session Hijacking** (e.g., compromised RDP/VPN session)
6. **Decision & Response Management**= Allow continued access if behavior is normal.
If anomaly is detected:
 - Trigger multi-factor authentication (MFA).
 - Temporarily lock or terminate the session.
 - Alert system administrators.
7. **Secure File System Integration**=Limit file access in the event that authentication is unsuccessful. Keep safe audit records of every anomaly and access attempt.

2. Non-Functional Requirements

1. **Performance** = The system must work in real time with little delay (less than one second for checks for problems). Shouldn't make the LAN file system work much less well.
2. **Scalability**=Be able to accommodate a number of users in various LAN settings. Capacity to modify ML models to accommodate various patterns of behavior.
3. **Accuracy**=Keep the accuracy of detection high (>90%) while reducing the number of false hits. Continuous improvement through learning with new information about users.
4. **Security**=Make sure that behavioral data is stored and sent securely. Do what the experts say about keeping your LAN safe and protecting users' privacy.
5. **Usability**= Offer a smooth experience to users with few interruptions. The prompts for secondary authentication should be easy to use.
6. **Reliability & Fault Tolerance**= The system must maintain operational functionality during LAN disruptions. Implement fail-safe mechanisms that enable legitimate users to reauthenticate in the event of false anomaly detection.

3. Hardware and Software Requirements

Hardware: A server that is capable of hosting machine learning models and has appropriate storage and processing capability. User computers, often known as LAN clients, that have monitoring agents attached to them. An architecture for networks that makes data collection and communication more secure.

Software

- Programming language: Python (for ML), C/C++ or Java (for LAN integration).
- Machine Learning frameworks: TensorFlow / PyTorch / Scikit-learn.
- Database: MySQL / PostgreSQL (to store behavioral data & profiles).
- Operating Systems: Windows/Linux (depending on LAN setup).
- Security libraries: For encryption, secure communication, and audit logging.

4. Data Requirements

- **Training Data:** User behavioral logs for supervised/semi-supervised ML model training.
- **Testing Data:** Simulated hijacking scenarios (physical + remote) to validate system performance.
- **Validation Data:** Continuous live data for model fine-tuning. [9] [10]

5. Personnel and Facilities

This research project will be conducted by a committed team under the leadership of the principal investigator, who will oversee the comprehensive design, development, and assessment of the SafeLAN system. The lead researcher will oversee tasks including performing the literature study, developing the machine learning model for behavioral anomaly detection, implementing the secure file system, and reporting the findings. The supervisor will closely oversee the work, offering academic and technical guidance to ensure the project retains both research depth and practical significance. The co-supervisor will offer further domain-specific insights, particularly in cybersecurity, secure storage techniques, and machine learning integration. Furthermore, colleagues and supportive team members may enhance the process through collaborative testing and feedback, thereby verifying the system's stability inside simulated LAN scenarios.

The resources necessary for the successful execution of this project will include both institutional and individual assets. The university laboratory will offer a regulated networking environment equipped with routers, switches, and servers, intended for the simulation of authentic LAN infrastructures for the deployment and testing of SafeLAN. These facilities will facilitate penetration testing and vulnerability assessments in a secure environment. The project will utilise programming environments including Python and C++, augmented by machine learning libraries and frameworks such as TensorFlow, PyTorch, and Scikit-learn. Security tools such as Wireshark, Nmap, and Metasploit will be utilized to assess network activity and system robustness, while databases like MySQL or PostgreSQL will facilitate the secure file management layer.

Alongside university infrastructure, individual resources will significantly contribute to the development process. A personal workstation with enough computational power, including a multi-core CPU and sufficient RAM, will be utilised for initial development and testing. Virtual machines will be designed to simulate isolated network environments, enabling secure testing without impacting actual systems. Moreover, access to institutional digital libraries and periodicals will furnish essential research materials, while cloud-based platforms may be employed for training the anomaly detection models if supplementary computational resources are needed.

The amalgamation of proficient individuals, supervisory oversight, and access to both physical and digital resources establishes a robust basis for the project's success and facilitates its effective execution within the designated timetable.

6. Budget

Category	Estimated Cost Range (LKR)
ML Model Development	Rs.5,000 – Rs.20,000
Authentication Platform	Rs.700 /user/month
Cloud Storage	Variable (based on usage)
On-Premises Infrastructure	Rs.10,000 – Rs.50,000+

7. Commercialization

Target Market

A wide variety of industries with different security requirements are catered to by SafeLAN. Since they sometimes have limited IT expenditures and need secure LAN file systems, small to medium-sized businesses (SMEs) make up a significant portion of the market. On the other hand, large organisations require constant surveillance to reduce insider threats because they handle valuable data. To protect sensitive information and secure internal communications, government and defence agencies require cutting-edge technologies. Additionally, because they must adhere to stringent data protection regimes like HIPAA, GDPR, and PCI DSS, regulated industries like healthcare and finance make up a significant portion of the target market.

Value Proposition

SafeLAN offers a special value because it combines high-level security with efficient operations. Its behaviour anomaly detection and ongoing authentication make it much less likely that insider threats or compromised accounts will happen. The system offers flexible monitoring that learns from and adapts to changing trends of user behaviour over time. SafeLAN's full logging and reporting features make sure that organisations with strict compliance standards are ready for audits. Also, automating anomaly spotting reduces the amount of work that IT teams have to do, so they don't have to do as much monitoring by hand and can focus on other important tasks.

Revenue Models

SafeLAN's revenue structure is designed to accommodate the diverse requirements of its clients by utilising a variety of revenue sources. A traditional software licensing model is available, which allows organisations to make a single purchase for on-premises deployment. The pricing is tiered and dependent on the number of users or endpoints. A subscription-based SaaS model provides cloud-managed services with monthly or yearly fees, which include updates, machine learning enhancements, and customer support, for organisations that are interested in flexibility. Consulting and customisation services are also available to enterprise clients, which facilitate the development of organization-specific ML models and the deployment of customised deployments. The revenue flow is further bolstered by support and maintenance services, which include training and compliance assistance.

Competitive Advantage

SafeLAN's competitive advantage is its ability to provide ML-based continuous authentication, which may detect small irregularities and insider threats that traditional systems often overlook.

Unlike conventional cloud-based security solutions, SafeLAN is designed exclusively for LAN situations. It seamlessly connects with current authentication systems, file servers, and IT infrastructures, ensuring adaptability across several organisations. Furthermore, it is cost-effective, decreasing the need for manual security activities while improving overall protection.

Marketing and Distribution

To increase SafeLAN's reach, a multifaceted marketing and distribution strategy is used. Direct sales focus on government and business customers who need sophisticated, specialised security solutions. Credibility and market penetration are increased by strategic alliances with cybersecurity companies and IT service providers. Online marketing showcases SafeLAN's distinctive ML-based continuous authentication to potential customers through webinars, whitepapers, and website demos. Additionally, attending industry conferences and cybersecurity events offers chances to present the solution to a worldwide expert audience.

Scaling and Expansion

One way that SafeLAN can grow in the long term is by following a plan for growing and expanding. The solution can be used in both LAN-based and cloud-based settings by integrating cloud services. It is planned to add more security features that will make it work better. These will include ransomware detection, encrypted backups, and role-based access control. The focus of SafeLAN will move towards global expansion as it grows, aiming at multinational companies and fields with strict compliance standards.

8. References

- [1] A. Adler and M. J. Mayhew, "IEEEExplore," 10 February 2014. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/6735899>.
- [2] S. P. Claudy Picard, "IEEEExplore," 14 June 2023. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/10151855>.
- [3] S. S. G. Y. Yunji Liang, "IEEEExplore," IEEE, 22 June 2020. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/9121981>.
- [4] S. Almalki, "mdpi," 30 June 2021. [Online]. Available: <https://www.mdpi.com/2076-3417/11/13/6083>.
- [5] Q. Y. Jiajia Li, "IEEEExplore," 14 October 2024. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/10716666>.
- [6] H. W. Run-Fa Liao, "mdpi.com," 28 May 2019. [Online]. Available: <https://www.mdpi.com/1424-8220/19/11/2440>.
- [7] B. K. M. E.-K. A. M. A. W. Kyle Quintal, "IEEEExplore," IEEE, 13 September 2019. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/8836081>.
- [8] S. H. S. H. F. M. ., P. P. Tarik Bin ShamsMd, "Research gate," April 2022. [Online]. Available: https://www.researchgate.net/publication/359789483_EEG-based_Biometric_Authentication_Using_Machine_Learning_A_Comprehensive_Survey?channel=doi&linkId=624eb1c9ef013420665f00d1&showFulltext=true.
- [9] R.-F. Liao, "mdpi," 28 May 2019. [Online]. Available: <https://www.mdpi.com/1424-8220/19/11/2440>.
- [10] A. O. Priya Bansal, "IEEEExplore.ie," IEEE, 15 August 2022. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/9851790>.

